

Arora V MSPI Configuration Mode Application Guide

Overview

In MSPI (Master SPI) configuration mode, the FPGA operates as an SPI master and actively reads the bitstream data from an external Flash device via the SPI interface to complete configuration.

The basic configuration flow is as follows:

1. Set the FPGA MODE pins to MSPI mode.
2. Power-cycle the FPGA, or apply a low-level pulse to the RECONFIG_N pin.
3. The FPGA automatically reads the bitstream data from the external Flash and completes the configuration.

Remote Upgrade Support

Supports remote upgrade via MSPI configuration. Supports writing new configuration data to external Flash remotely during FPGA operation. Upgrade completed by RECONFIG_N trigger or power cycle at the proper time.

Supported Modes

MSPI supports the following three SPI data line modes:

- X1 (Single) – Single data line mode.
- X2 (Dual) – Dual data line mode.
- X4 (Quad) – Quad data line mode.

Additional Features

- Multiboot
- Golden image

Multiboot, a derived concept of the MSPI configuration mode, refers to the process where the FPGA reads bitstream data from different addresses of the external Flash to perform configuration. In Multiboot operation, the user embeds the load address of the next bitstream into the current bitstream. While the device remains powered, the FPGA can switch to the

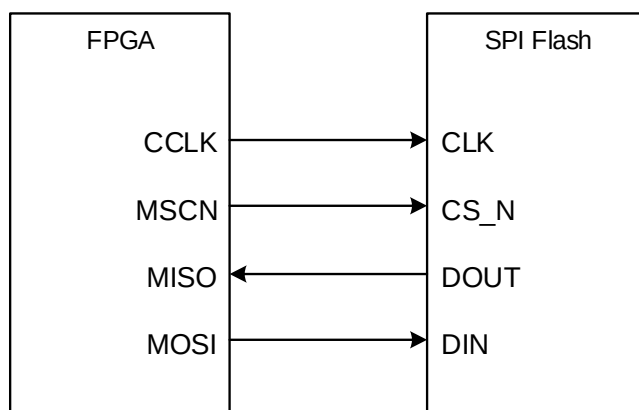
new bitstream by triggering the RECONFIG_N pin or issuing a reboot command to complete reconfiguration. FPGA products that supports MSPI all support this mode.

The Golden Image refers to a predefined backup bitstream. If an error occurs during normal bitstream loading, the device will automatically load the Golden Bitstream from the preset Golden Address (0x800000) to ensure system operation.

Connection Diagram for MSPI Configuration Mode

The connection diagram for configuring Gowin FPGA products using MSPI x1/x2/x4 configuration modes is shown in Figure 1 ~ Figure3.

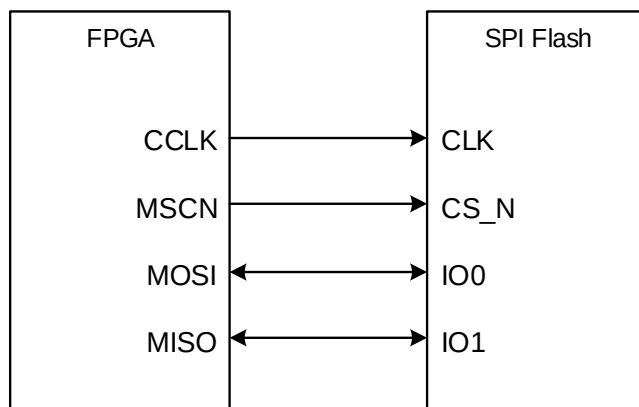
Figure 1 Connection Diagram for MSPIx1 Configuration Mode



Note!

- MSPI x1 frequency range: 2.5Mhz~105Mhz
- The default configuration is MSPI x1 with 3-byte addressing, which can be modified in the EDA software to reduce the loading time.

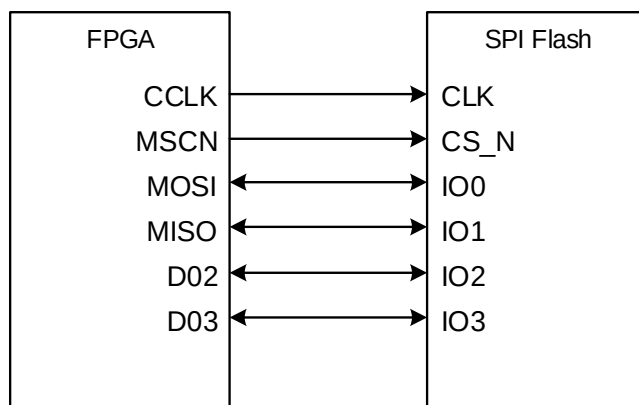
Figure2 Connection Diagram for MSPIx2 Configuration Mode



Note!

- MSPI x2 frequency range: 2.5Mhz~105Mhz
- The default configuration is MSPI x1 with 3-byte addressing, which can be modified in the EDA software to reduce the loading time.

Figure3 Connection Diagram for MSPIx4 Configuration Mode



Note!

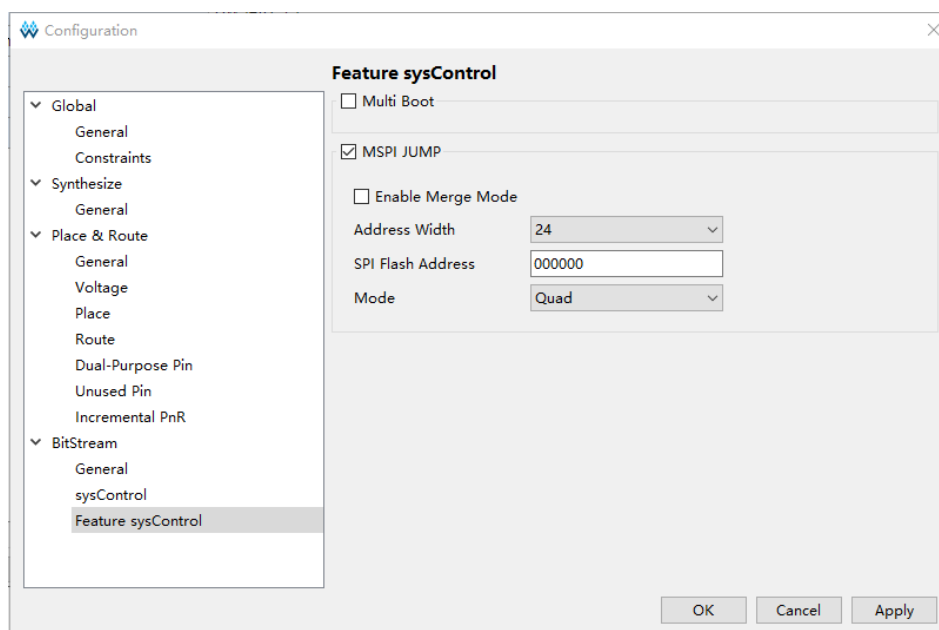
- COMPONENT x4 frequency range: 2.5Mhz~105Mhz
- The default configuration is MSPI x1 with 3-byte addressing, which can be modified in the EDA software to reduce the loading time.

MSPI Jump

In MSPI mode, by using a jump file and inserting the instruction `mspi_jump(0x6E)` at the beginning of the bitstream, the FPGA can jump to an arbitrary load address. This enables flexible allocation of Flash storage space in complex systems.

- The jump file allows the FPGA to load the bitstream from any arbitrary address. `jump.fs` must be stored at Flash address 0x0 to use the jump function.
- MSPI Quad/Dual modes require a jump file, e.g., for loading the Golden Image or bitstream at Flash address 0x0.
- The jump file can be a standalone bitstream or merged with the user's functional bitstream into a single file.

The jump file can be generated as follows. The SPI Flash Address specifies the target load address for the jump.



For a specific example, please refer to MSPI Quad Mode (X4) Application > 4 Using the MSPI Jump File for Address Jumping.

Application Example and Operation Flow

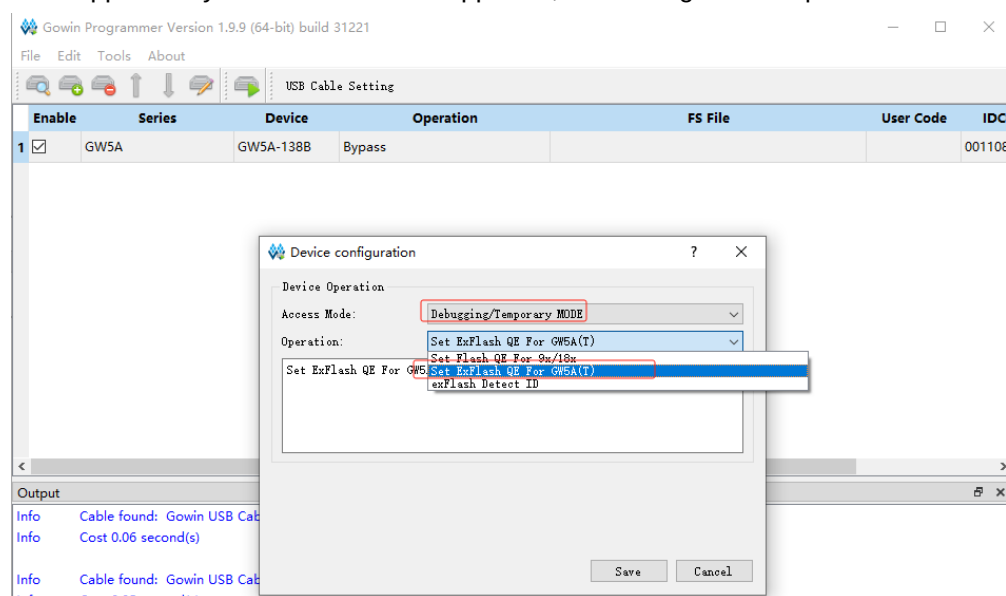
QE (Quad Enable bit) Enable

To use MSPI Quad mode, the QE bit of the external SPI Flash must be enabled first.

- Procedure: Use the Programmer tool to program the Flash and set its QE bit, as shown in the figure below. This operation only needs to be performed once and retains its setting after power cycling.

Note!

Please refer to the related datasheet to confirm whether Quad mode is supported by default. If it is not supported, this setting is not required.



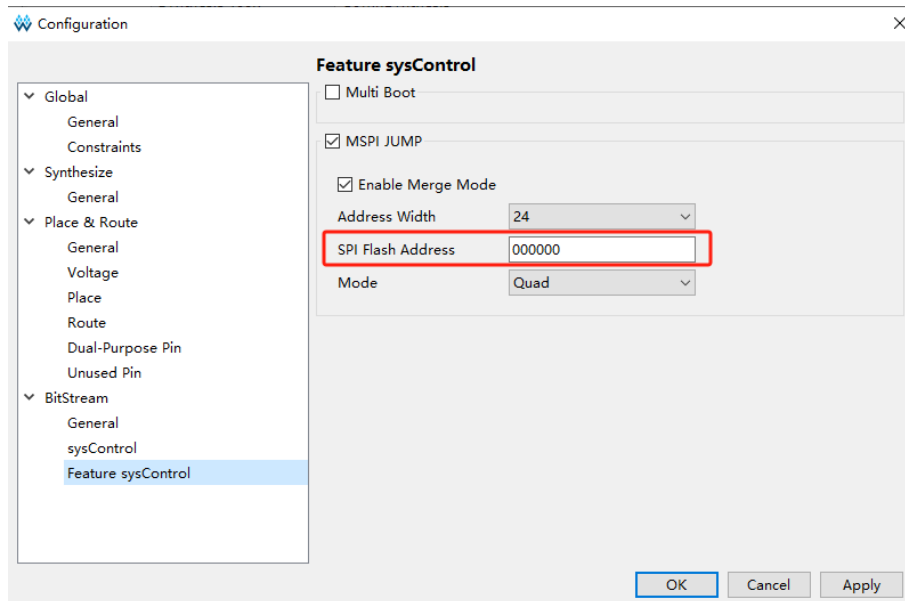
MSPI Quad Mode (X4) Application

In Quad mode, ensure that the QE (Quad Enable) bit of the SPI Flash is set.

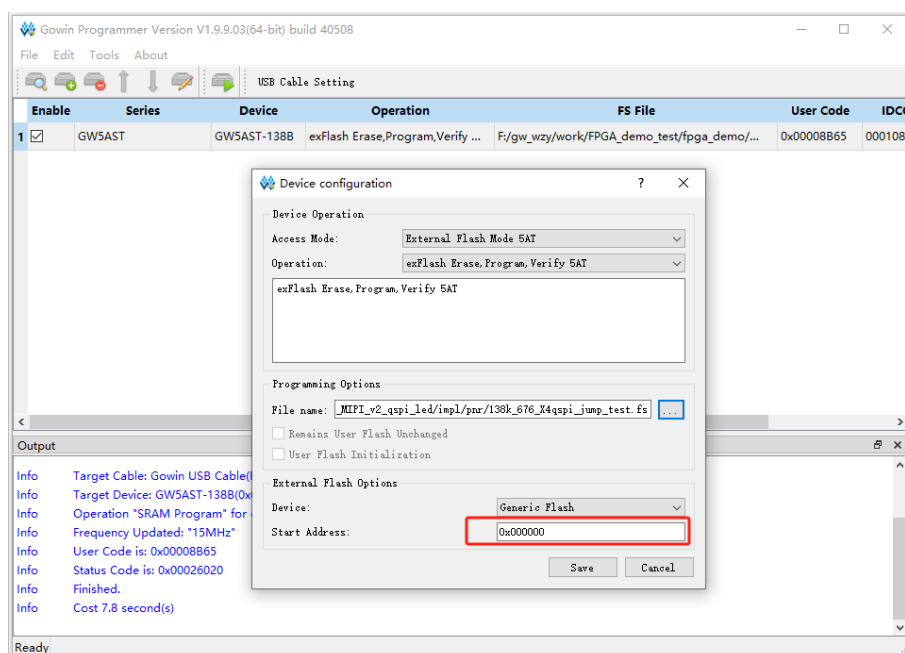
1. Single Bitstream Loading

1) Bitstream Feature Configuration in Gowin Software

It is recommended to check “Enable Merge Mode.” Selecting this option will merge the MSPI JUMP bitstream into the user’s main bitstream.



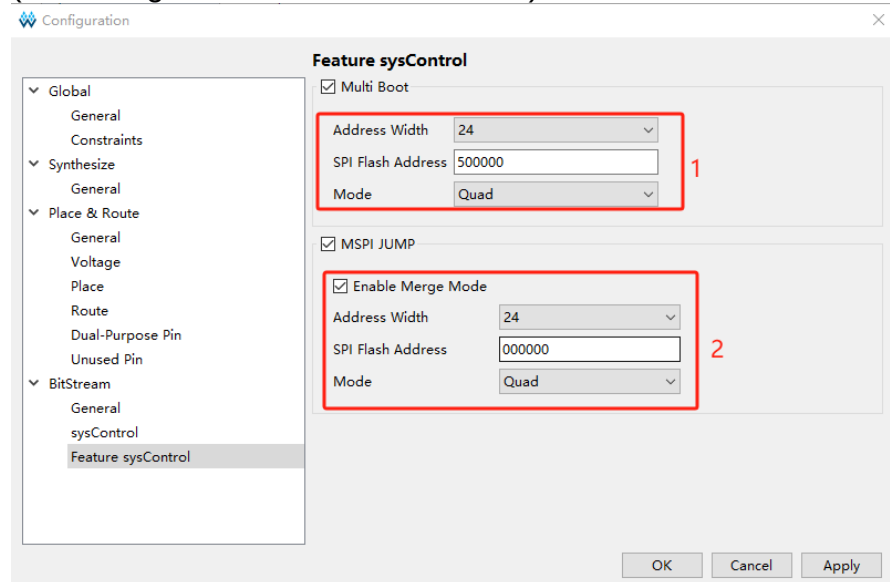
2) Download the bitstream to address 0x0 of the Flash using the Programmer tool.



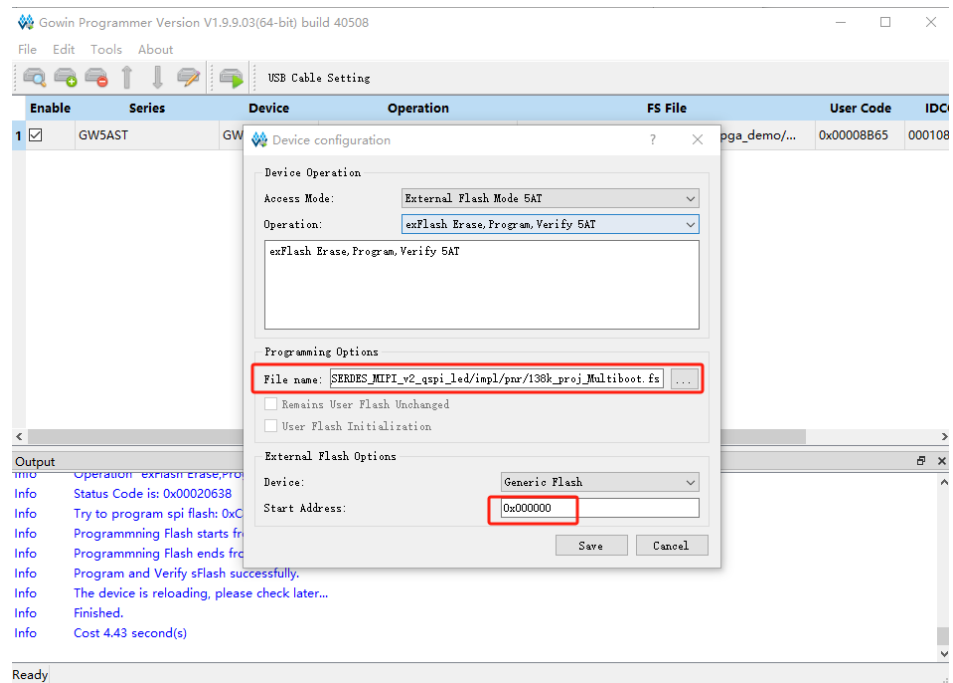
2. Multiboot

1) Bitstream Configuration in Gowin Software (Generate FS1)

- Select “Multi Boot” and set the next jump address to 0x500000.
- Select “MSPI JUMP” and set the jump address to 0x000000 (the storage address of FS1 in Flash).



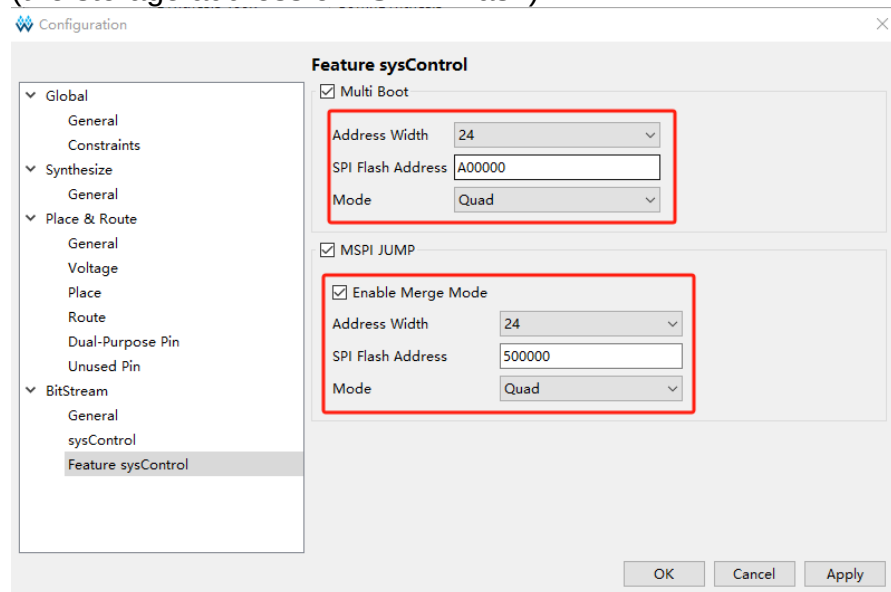
2) Download the FS1 bitstream to address 0x00 of the Flash using the Programmer tool.



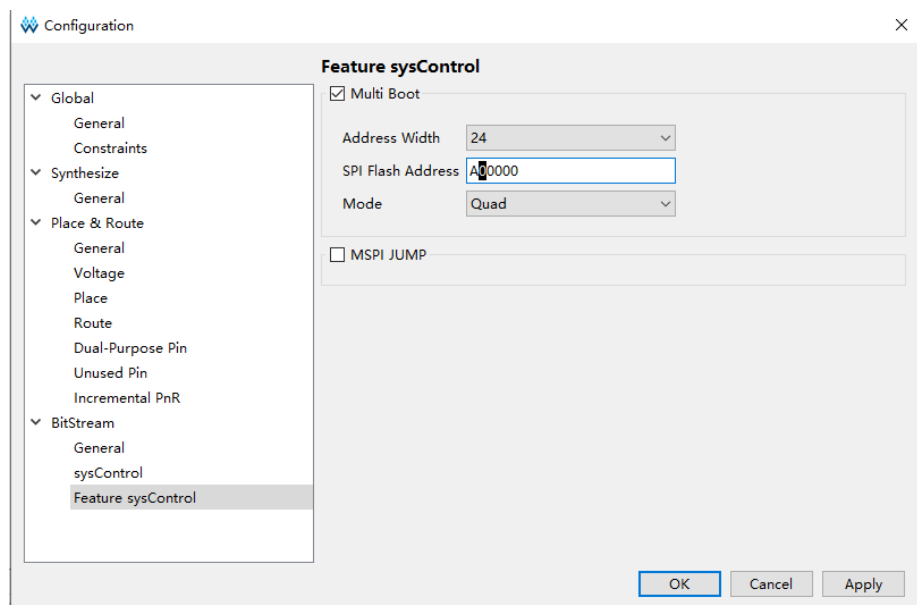
3) Bitstream Configuration in Gowin Software (Generate FS2)

- Select “Multi Boot” and set the next jump address to 0xA00000.

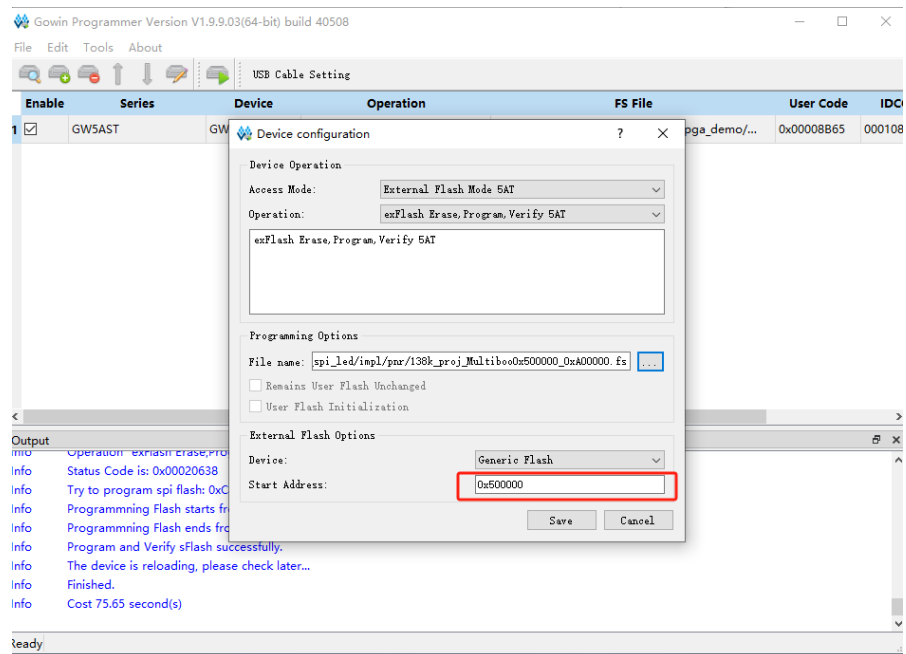
- Select “MSPI JUMP” and set the jump address to 0x500000 (the storage address of FS2 in Flash).



Bitstream 2 can also be configured via the “Configuration” settings, as shown in the figure below.

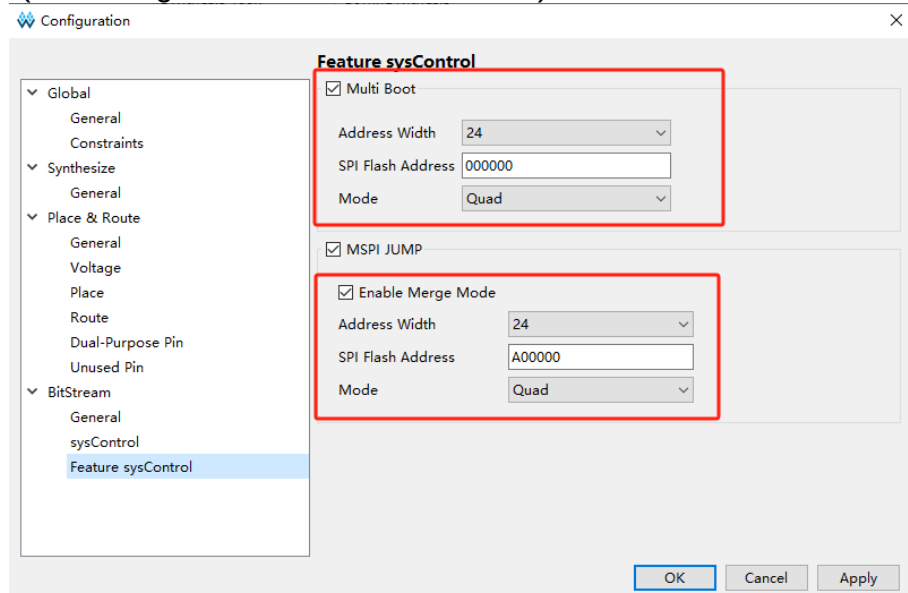


- 4) Download the FS2 bitstream to address 0x500000 of the Flash using the Programmer tool.

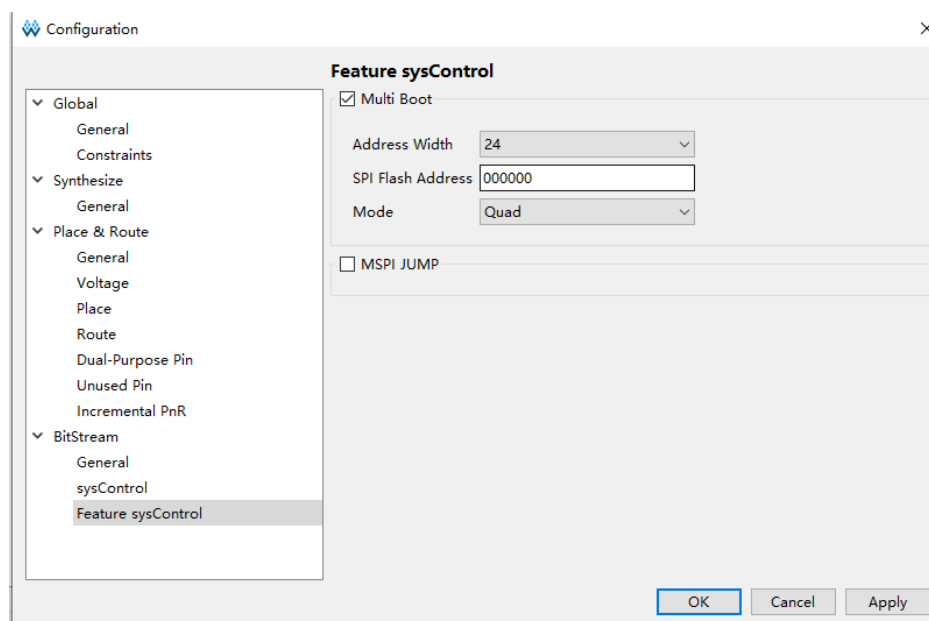


5) Bitstream Configuration in Gowin Software (Generate FS3)

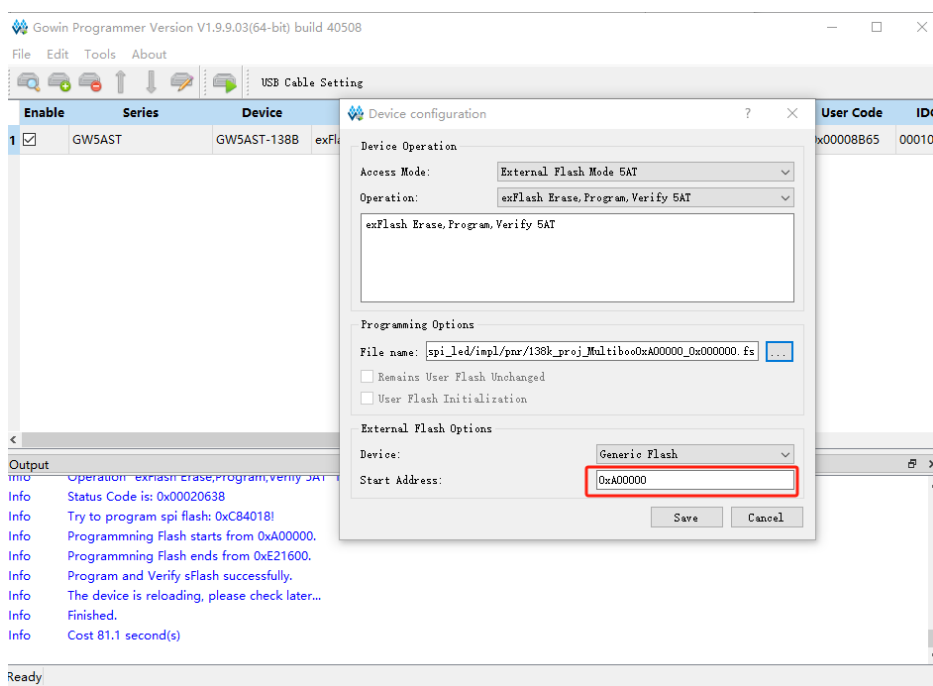
- Select “Multi Boot” and set the next jump address to 0x000000.
- Select “MSPI JUMP” and set the jump address to 0xA00000 (the storage address of FS3 in Flash).



Bitstream 3 can also be configured via the “Configuration” settings, as shown in the figure below.

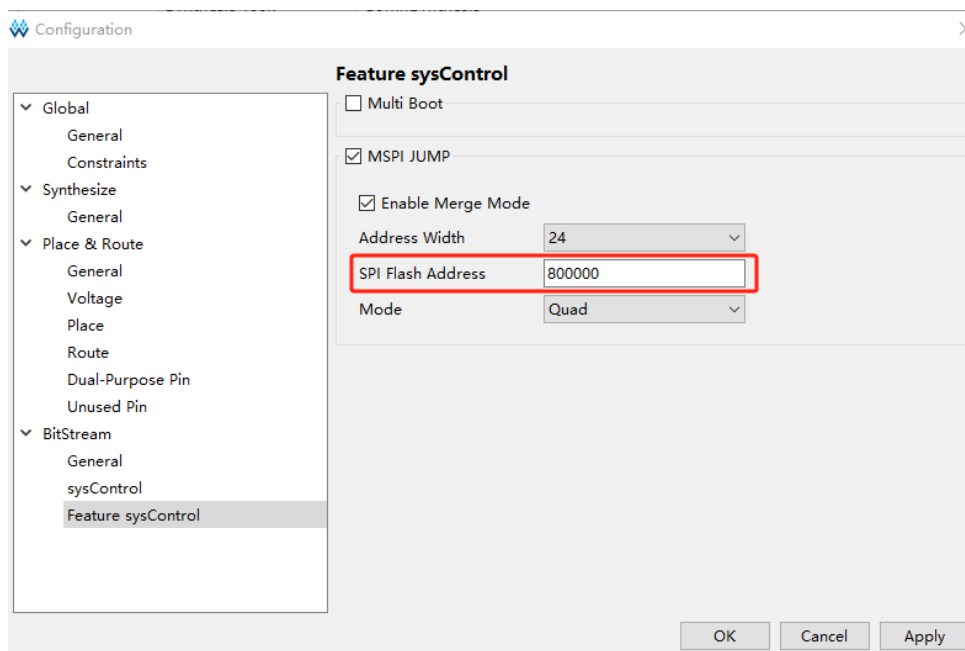


- 6) Download the FS3 bitstream to address 0xA00000 of the Flash using the Programmer tool.

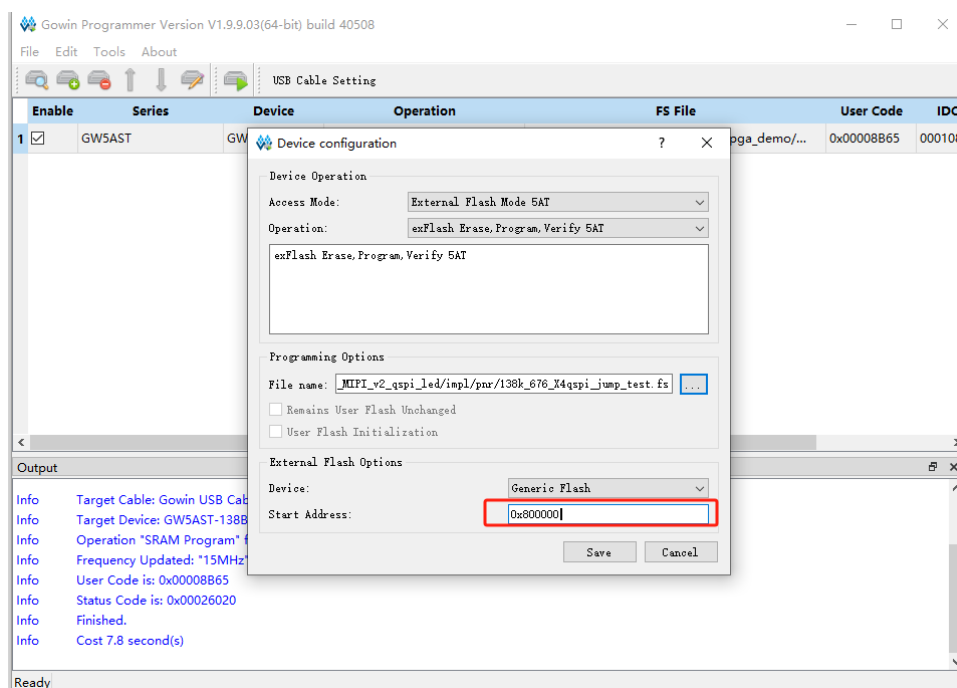


- 7) Triggering the RECONFIG_N pin enables the Multiboot function.
3. Golden image
- 1) Bitstream Configuration in Gowin Software

It is recommended to check “Enable Merge Mode.” Selecting this option will merge the MSPI JUMP bitstream into the user’s main bitstream.



- 2) Download the FS2 bitstream to address 0x800000 of the Flash using the Programmer tool.



Note!

Ensure that the SPI Flash has a capacity larger than 8 MB (e.g., 25Q64). If the Flash is smaller than 8 MB, the Golden Image address must be adjusted accordingly.

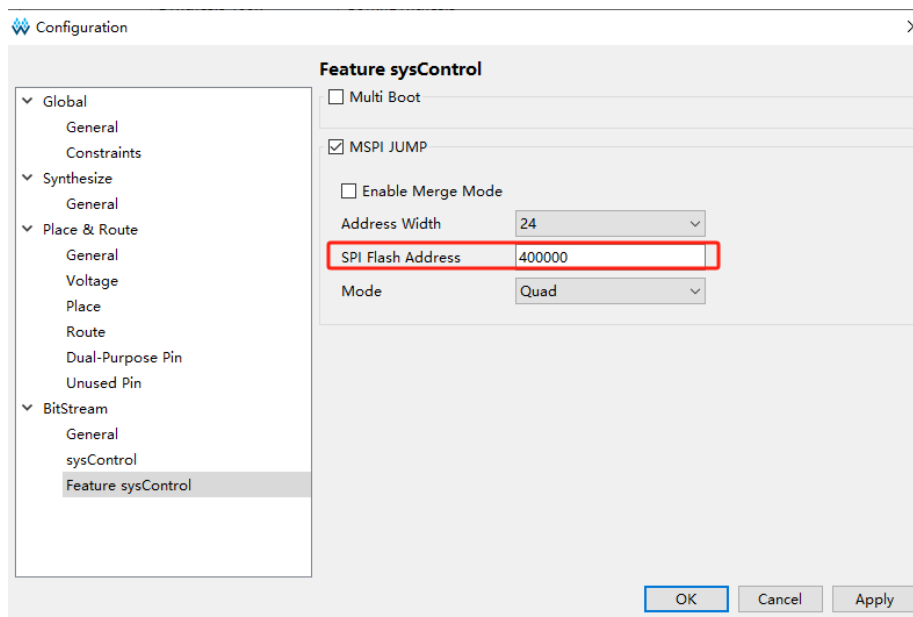
- 3) Golden Image Loading

The FPGA first loads the bitstream from address 0x0 of the Flash. If the bitstream at 0x0 is corrupted, it automatically jumps to address 0x800000 to load the Golden Image.

4. Using the MSPI Jump File for Address Jumping

1) Configuring MSPI JUMP in Gowin Software

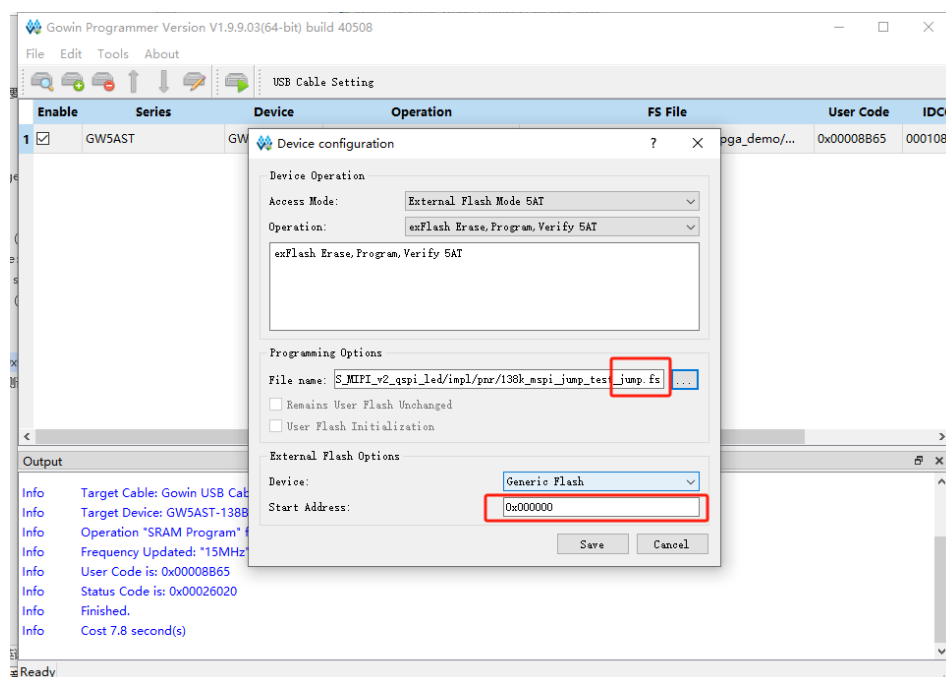
Set the SPI Flash Address to 0x400000. This address is used as the jump target and corresponds to the starting address of the user bitstream loaded by MSPI.



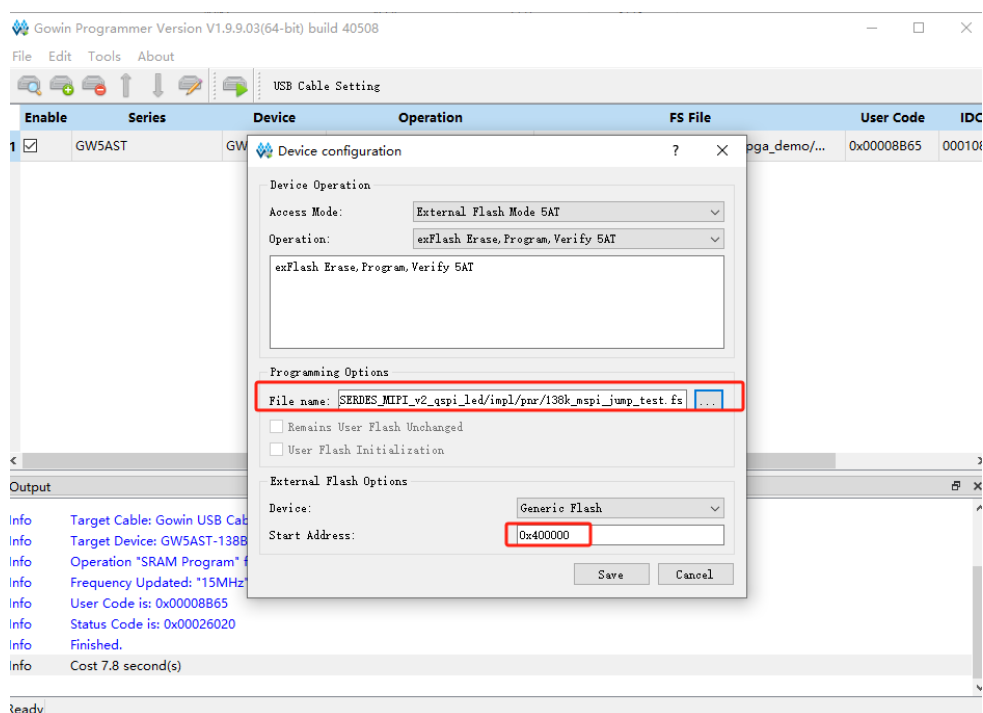
After placement and routing (PNR), two FS files are generated in the PNR output directory. The jump.fs file has a size of 2 KB.

FPGA_demo_test > fpga_demo > GW5AT > 138K > 138K_PG676_SERDES_MIPI_v2_qspi_led > impl > pnr				
名称	修改日期	类型	大小	
138k_msipi_jump_test.pin.html	2024/6/6 9:30	360 Chrome HTML Docume...	86 KB	
138k_msipi_jump_test.rpt.html	2024/6/6 9:30	360 Chrome HTML Docume...	91 KB	
138k_msipi_jump_test.tr.html	2024/6/6 9:30	360 Chrome HTML Docume...	1 KB	
138k_msipi_jump_test_tr_cata.html	2024/6/6 9:30	360 Chrome HTML Docume...	8 KB	
138k_msipi_jump_test_tr_content.html	2024/6/6 9:30	360 Chrome HTML Docume...	236 KB	
138k_msipi_jump_test.bin	2024/6/6 9:30	BIN 文件	4,230 KB	
138k_msipi_jump_test_jump.bin	2024/6/6 9:30	BIN 文件	1 KB	
138k_msipi_jump_test.binx	2024/6/6 9:30	BINX 文件	4,230 KB	
138k_msipi_jump_test_jump.binx	2024/6/6 9:30	BINX 文件	1 KB	
device.cfg	2024/6/6 9:30	Configuration 源文件	1 KB	
138k_msipi_jump_test.db	2024/6/6 9:30	Data Base File	2 KB	
cmd.do	2024/6/6 9:30	DO 文件	1 KB	
138k_msipi_jump_test.fs	2024/6/6 9:30	F# 源文件	33,878 KB	
138k_msipi_jump_test_jump.fs	2024/6/6 9:30	F# 源文件	2 KB	
138k_msipi_jump_test.timing_paths	2024/6/6 9:30	TIMING_PATHS 文件	8 KB	
138k_msipi_jump_test.log	2024/6/6 9:30	文本文档	2 KB	
138k_msipi_jump_test.rpt.txt	2024/6/6 9:30	文本文档	69 KB	

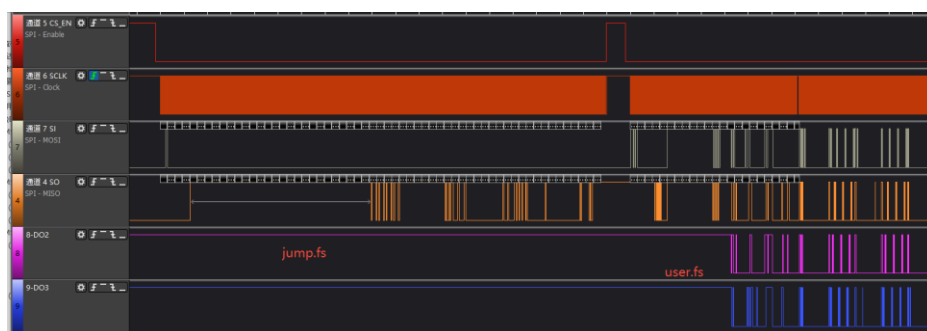
2) Download the jump.fs file to address 0x00 of the Flash using the Programmer tool.



- 3) Download the user bitstream with functionality to address 0x400000 of the Flash using the Programmer tool.



- 4) MSPI Loading Timing on Power Re-up

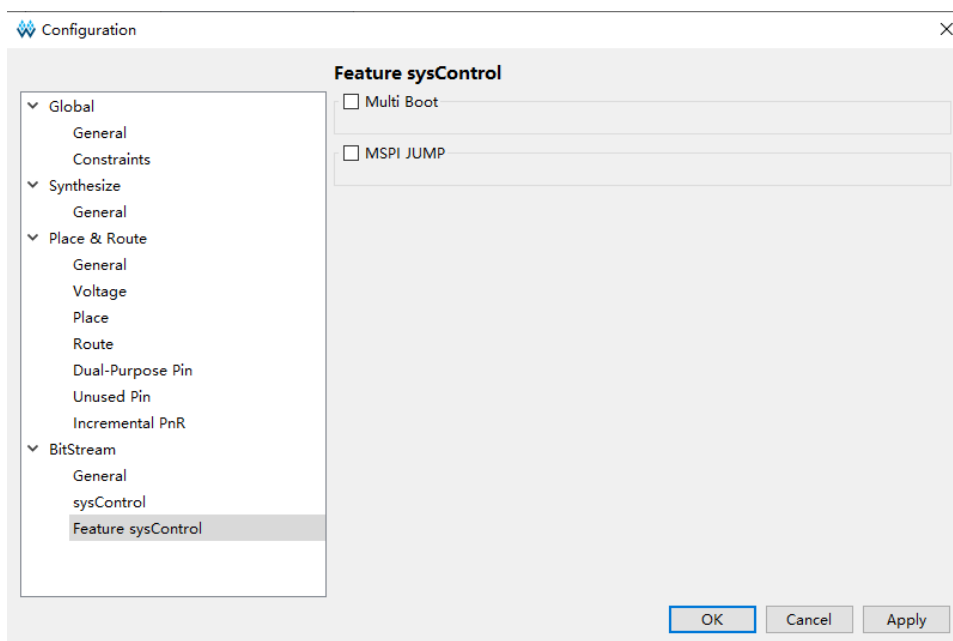


MSPI Single (X1) Application

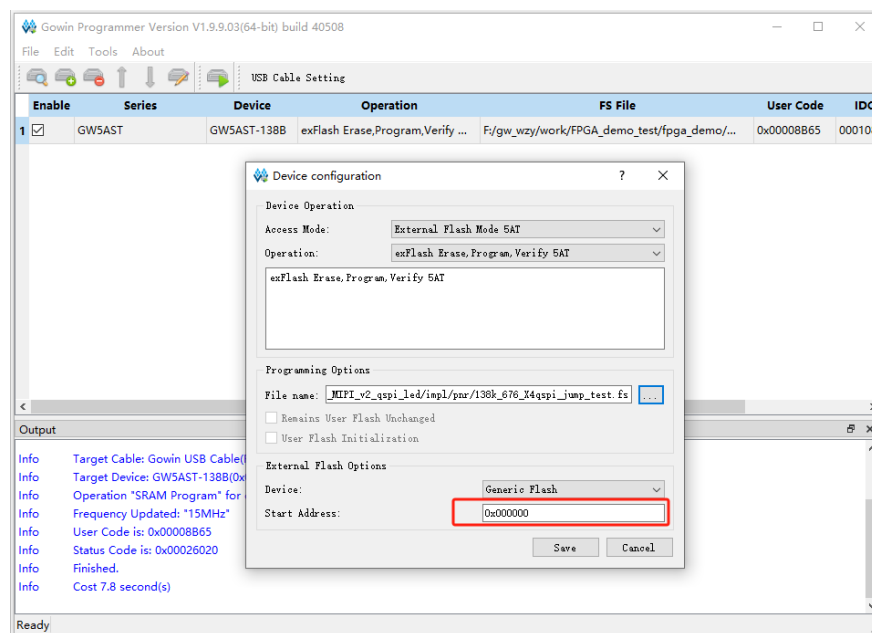
1. Loading a Single Bitstream

1) Bitstream Configuration in Gowin Software

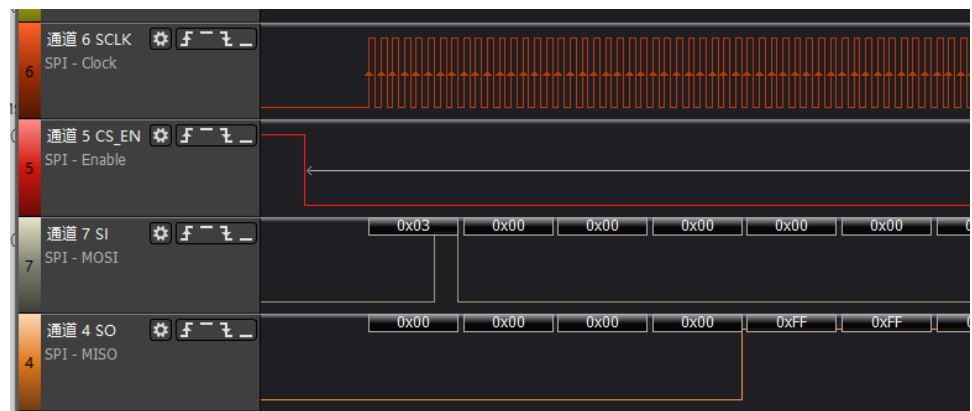
Do not check “MSPI JUMP” or “Multi Boot” options



2) Download the Bitstream to Flash address 0x0 using the Programmer tool.



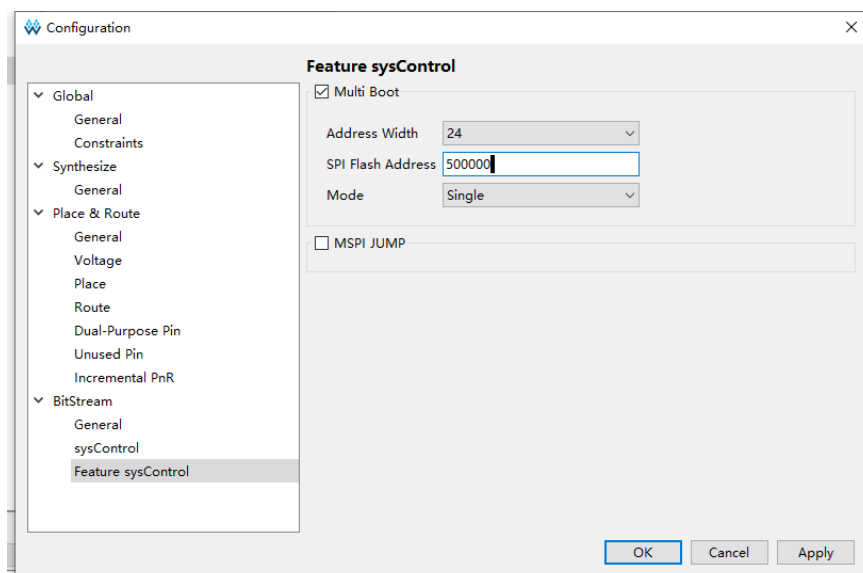
3) MSPI Loading Timing on Power Re-up



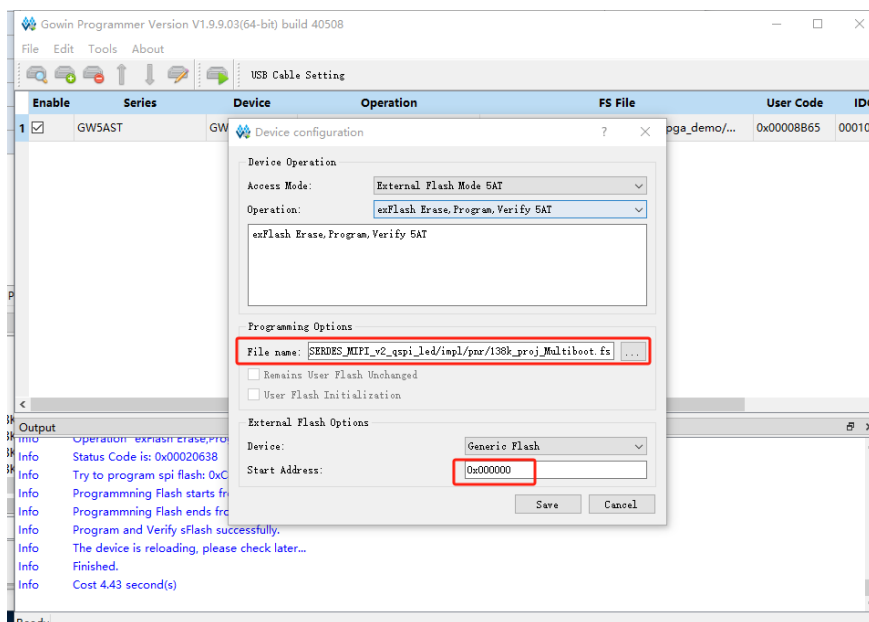
2. Multiboot

1) Bitstream Configuration in Gowin Software (Generate FS1)

Check the “Multi Boot” option and set the next jump address to 0x500000.

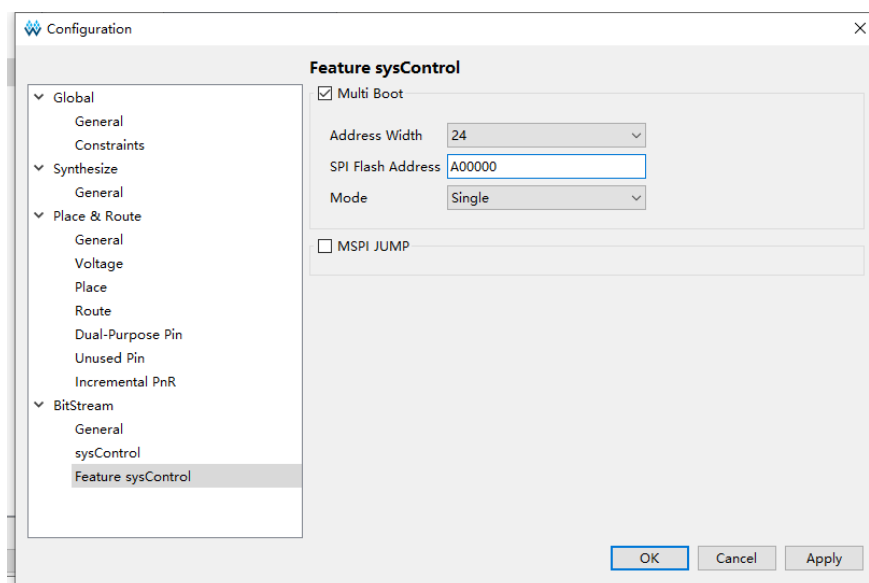


2) Download the FS1 bitstream to address 0x00 of the Flash using the Programmer tool.

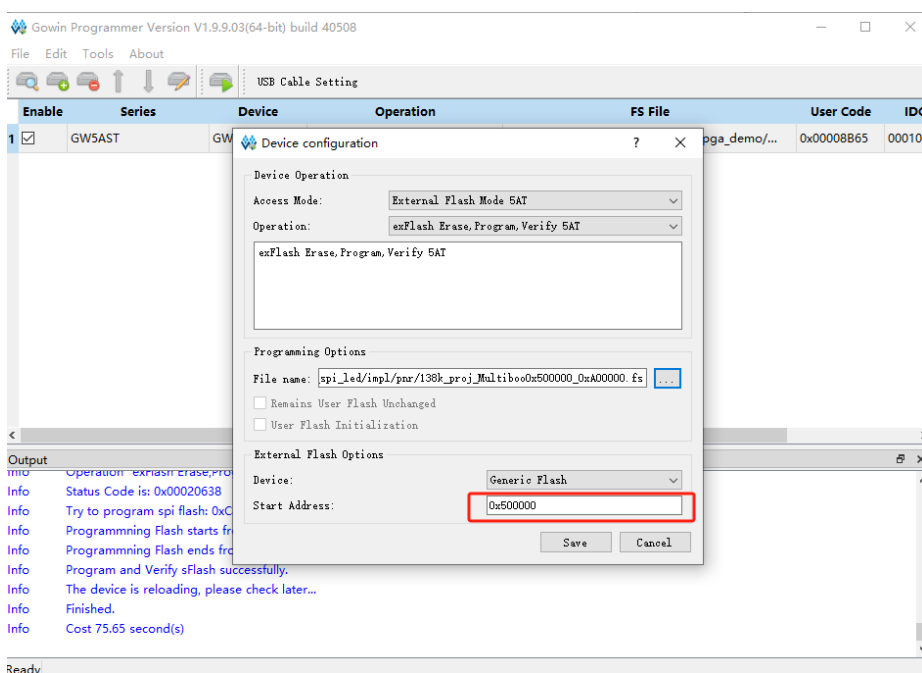


3) Bitstream Configuration in Gowin Software (Generate FS2)

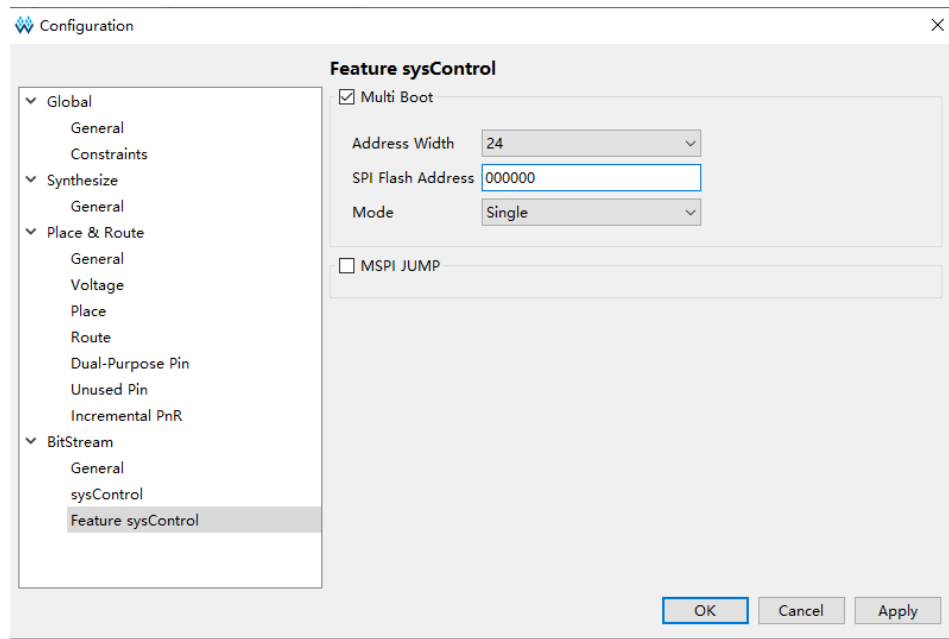
Check the “Multi Boot” option and set the next jump address to 0xA00000.



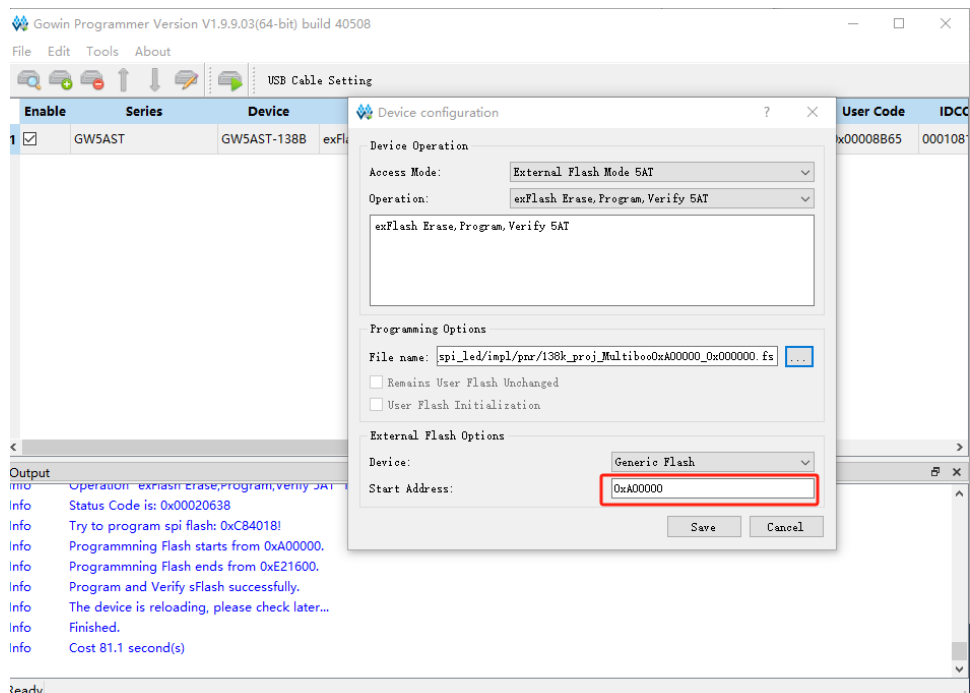
- 4) Download the FS2 bitstream to address 0x500000 of the Flash using the Programmer tool.



- 5) Bitstream Configuration in Gowin Software (Generate FS3)
Check the “Multi Boot” option and set the next jump address to 0x000000.



- 6) Download the FS3 bitstream to address 0xA00000 of the Flash using the Programmer tool.

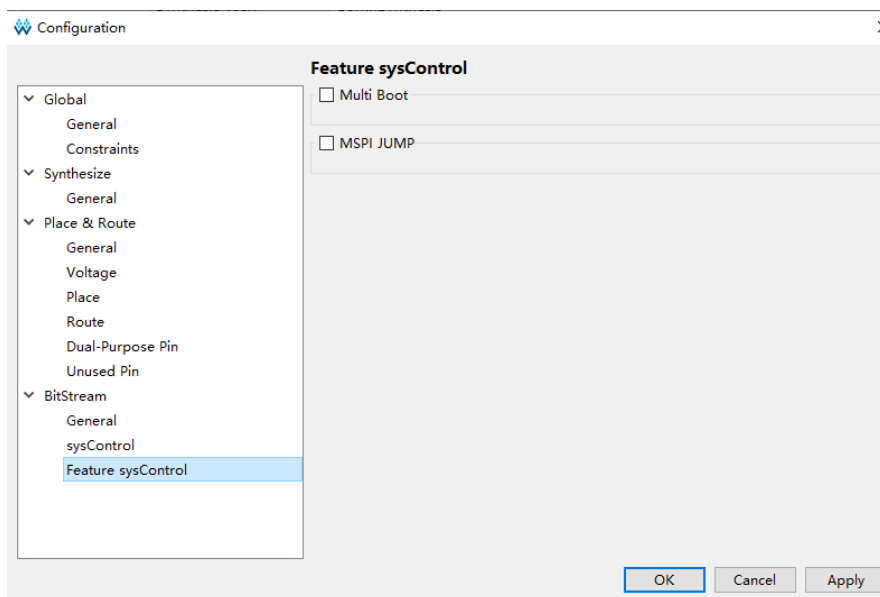


- 7) Triggering the RECONFIG_N pin enables the Multiboot function.

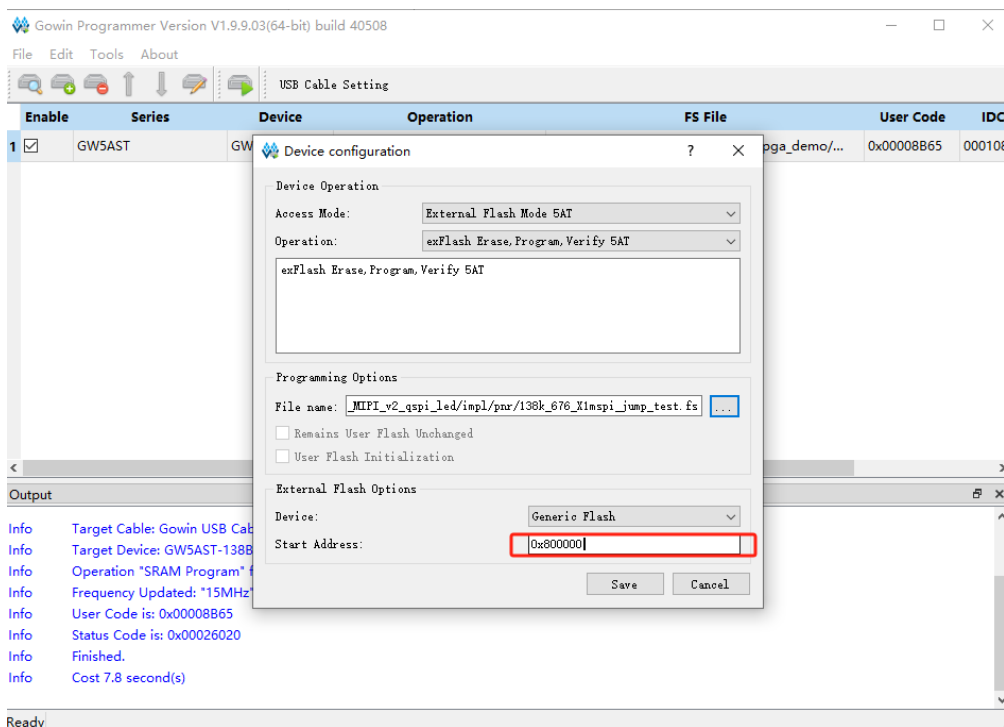
3. Golden image

- 1) Bitstream Configuration in Gowin Software

Do not check “MSPI JUMP” or “Multi Boot” options

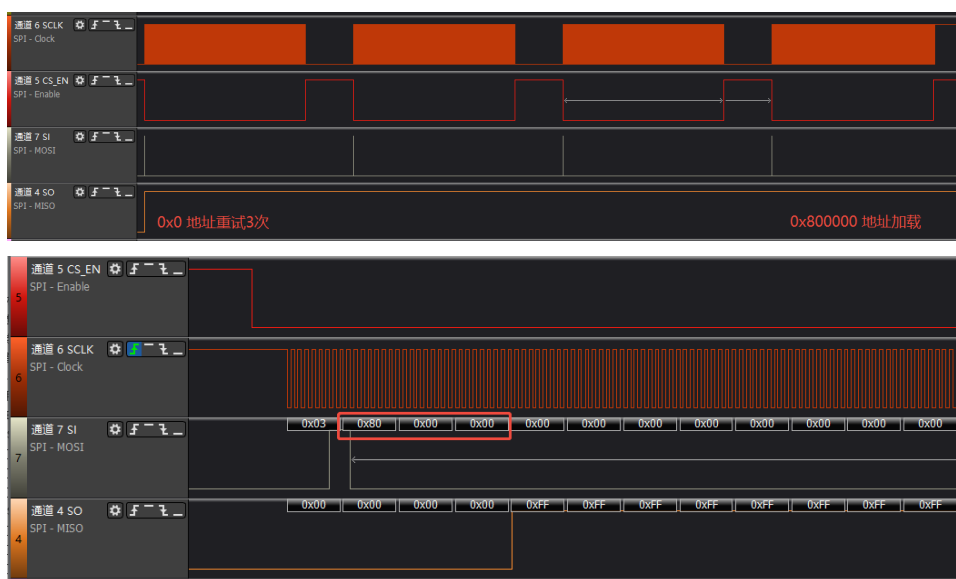


- 2) Download the Bitstream to address 0x8000000 of the Flash using the Programmer tool.



- 3) MSPI Loading Timing on Power Re-up

The FPGA first loads the bitstream from address 0x0 of the Flash. If the bitstream at 0x0 is corrupted, it automatically jumps to address 0x800000 to load the Golden Image.



MSPI Single (X2) Application

The operation flow and configuration logic for Dual mode (X2) are essentially the same as for Quad mode (X4). Please refer to [MSPI Quad Mode \(X4\) Application](#) for configuration and operation.

Notes

1. By default, upon power-up the FPGA loads its configuration starting from Flash address 0x0 using MSPI Single (X1) mode.
2. Mode consistency: If the bitstream stored at Flash address 0x0 is configured for Quad mode and 24-bit address width, the corresponding Golden Image or Multiboot image must also be configured with Quad mode and 24-bit address width to ensure proper loading.
3. Load fault tolerance: During bitstream loading, if a verification error occurs, the FPGA will typically attempt to reload the bitstream up to three times. If all three attempts fail, the error handling mechanism is triggered and the FPGA loads the Golden Image.
4. Clock frequency: The MSPI clock supports frequencies up to 105 MHz. Before using this high-speed clock, please ensure that the following conditions are met:
 - Flash support: Confirm that the chosen SPI Flash device can operate reliably at the configured MSPI frequency.
 - PCB and signal integrity: Ensure PCB design meets signal integrity requirements. It is recommended to length-match SPI clock and data signals to minimize timing skew.
5. Flash storage capacity: When using Multiboot or Golden Image configurations, plan the Flash address space in advance to ensure sufficient capacity and prevent bitstream overwriting.

Applicable Scenarios and Application Summary

The MSPI configuration mode described in this document applies to the following Gowin Arora V FPGA devices:

- GW5AT-138 / GW5AT-75 / GW5AT-60 / GW5AT-15
- GW5A-138 / GW5A-25/ GW5A-60
- GW5AR-25
- GW5ART-15
- GW5AS-25/GW5AS-138
- GW5AST-138

Key applicable scenarios:

1. **Single Bitstream Loading:** The most basic configuration method, suitable for scenarios where functionality is stable.
2. **Multiboot:** Supports switching between multiple configuration images, suitable for functionality upgrades and multi-mode operation scenarios.
3. **Golden Image:** Ensures system recovery and basic operation.
4. **Remote Upgrade:** Enables system configuration updates through a communication interface, allowing remote FPGA bitstream updates.
5. **Jump Address Switching:** Provides flexible management of Flash storage space and supports complex memory layouts.

Support and Feedback

Gowin Semiconductor provides customers with comprehensive technical support. If you have any questions, comments, or suggestions, please feel free to contact us directly using the information provided below.

Website: www.gowinsemi.com

E-mail: support@gowinsemi.com

Revision History

Date	Version	Description
2026/01/12	1.0E	Initial version published.

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